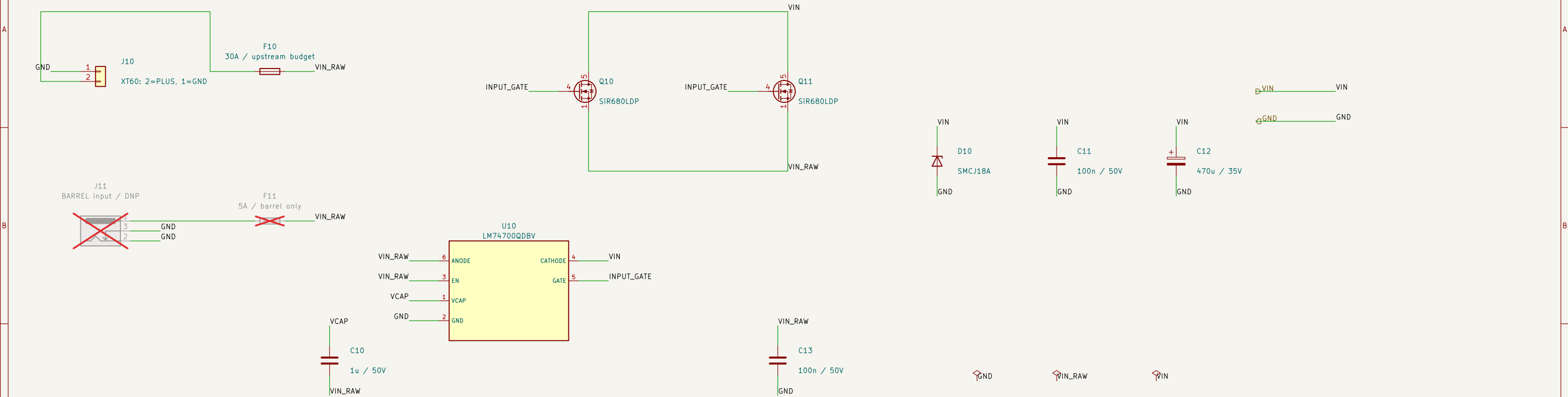


PULSARPOWER | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN – routing, full-load thermal validation and new firmware are still required.

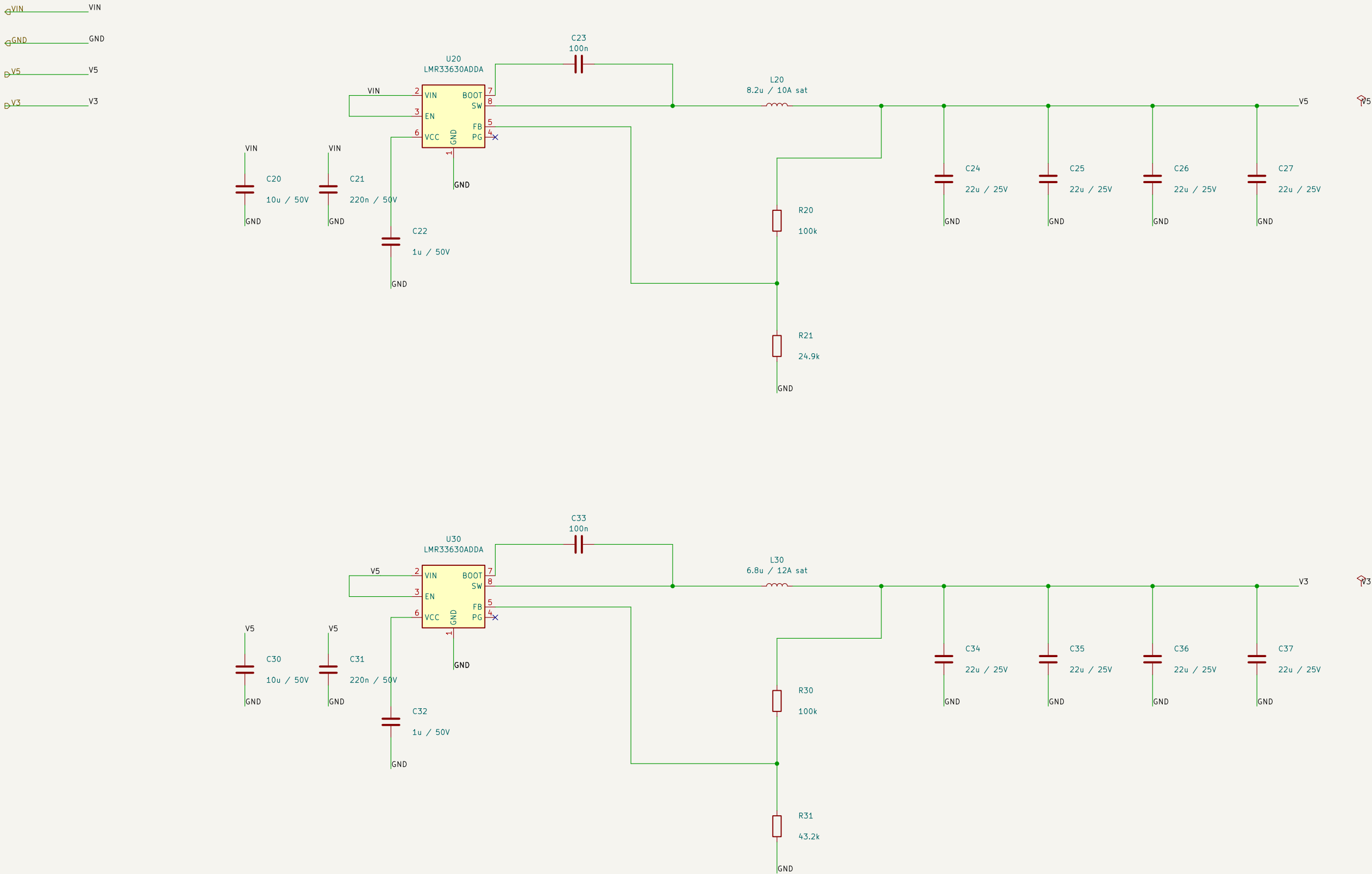


POWER INPUT | 12–18 V nominal | XT60 or barrel assembly option



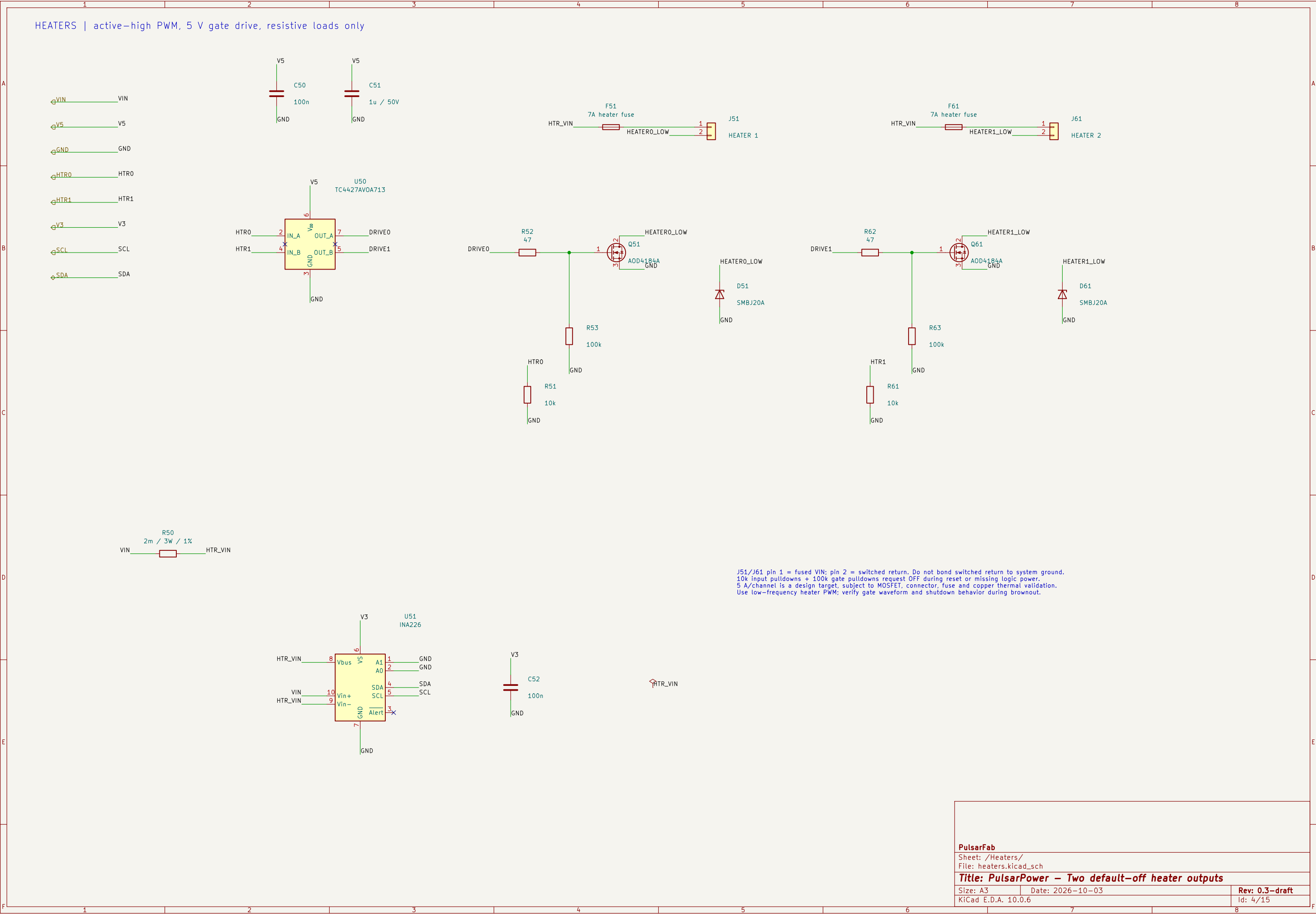
FIT J10 + F10 OR J11 + F11. Do not populate both input connector paths.
XT60: provisional 25 A shared continuous budget; fuse, copper and temperature still require validation.
Barrel: 5 A absolute TOTAL ceiling; 3.5 A provisional continuous budget after fuse derating. Use a current-limited supply.
LM74700 + parallel Q10/Q11 provide reverse-polarity / reverse-current protection. It is not an overvoltage disconnect.
Never apply more than 18 V nominal. TVS pulse capability is not a sustained overvoltage rating.

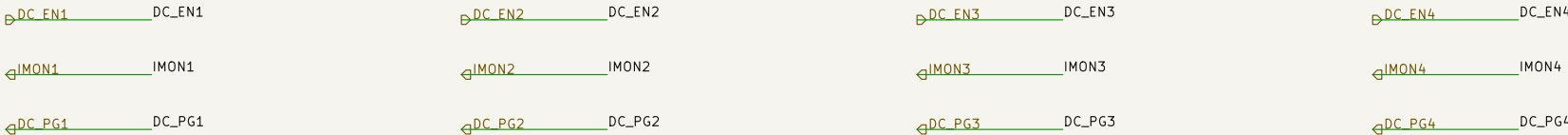
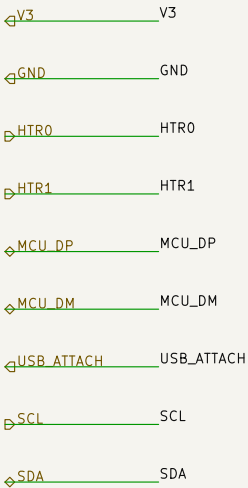
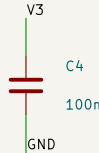
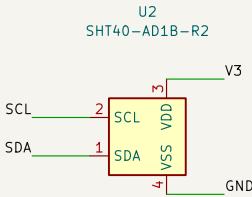
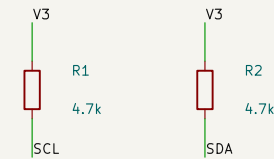
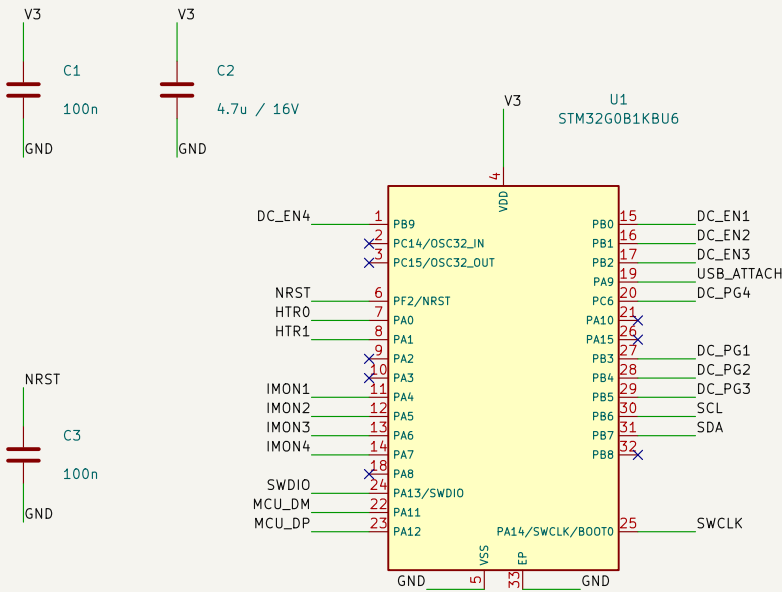
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic



Follow TI SNVSN3F Table 9–2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus 0.6 A reserved for onboard logic/support. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

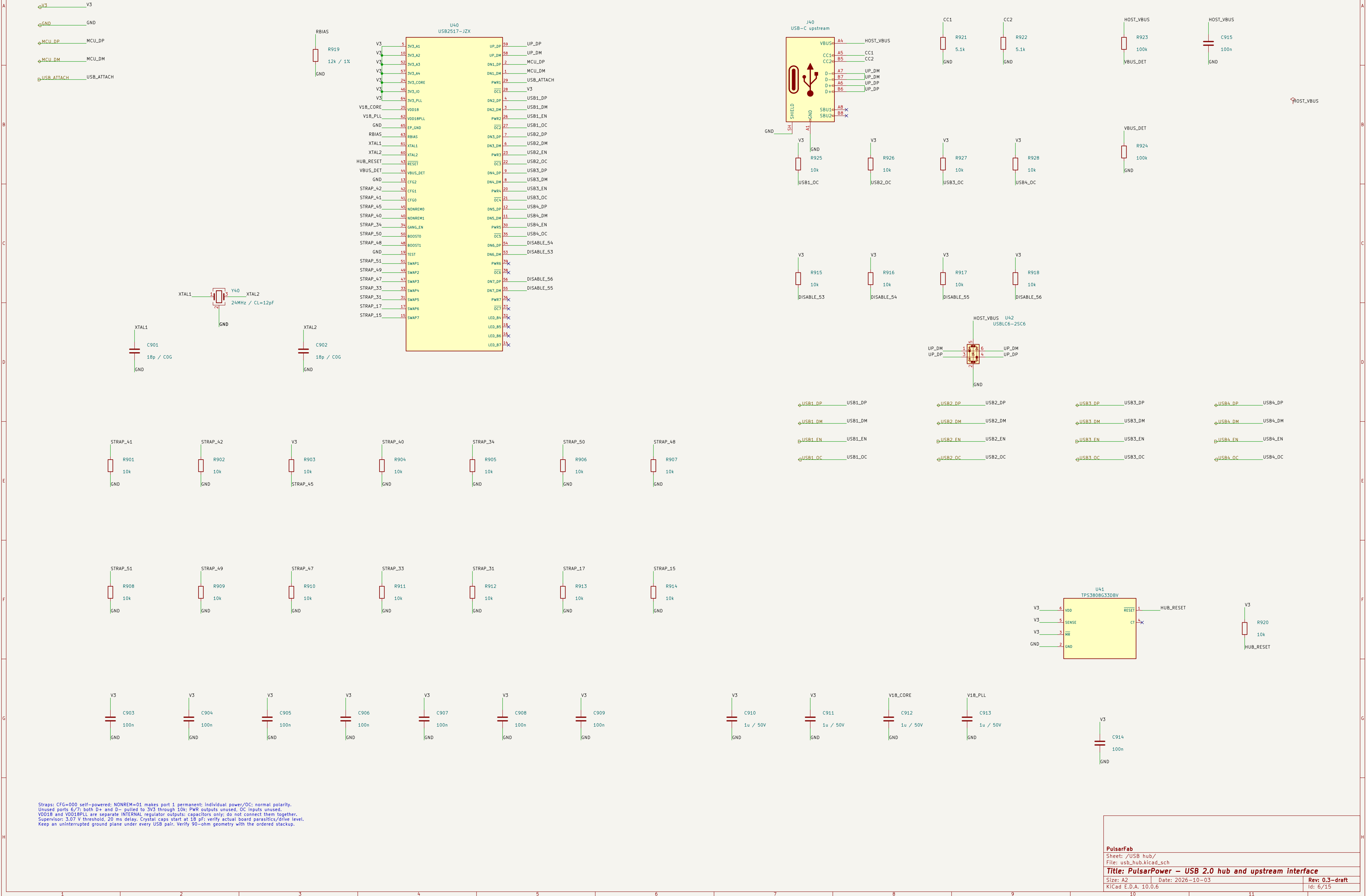
HEATERS | active-high PWM, 5 V gate drive, resistive loads only



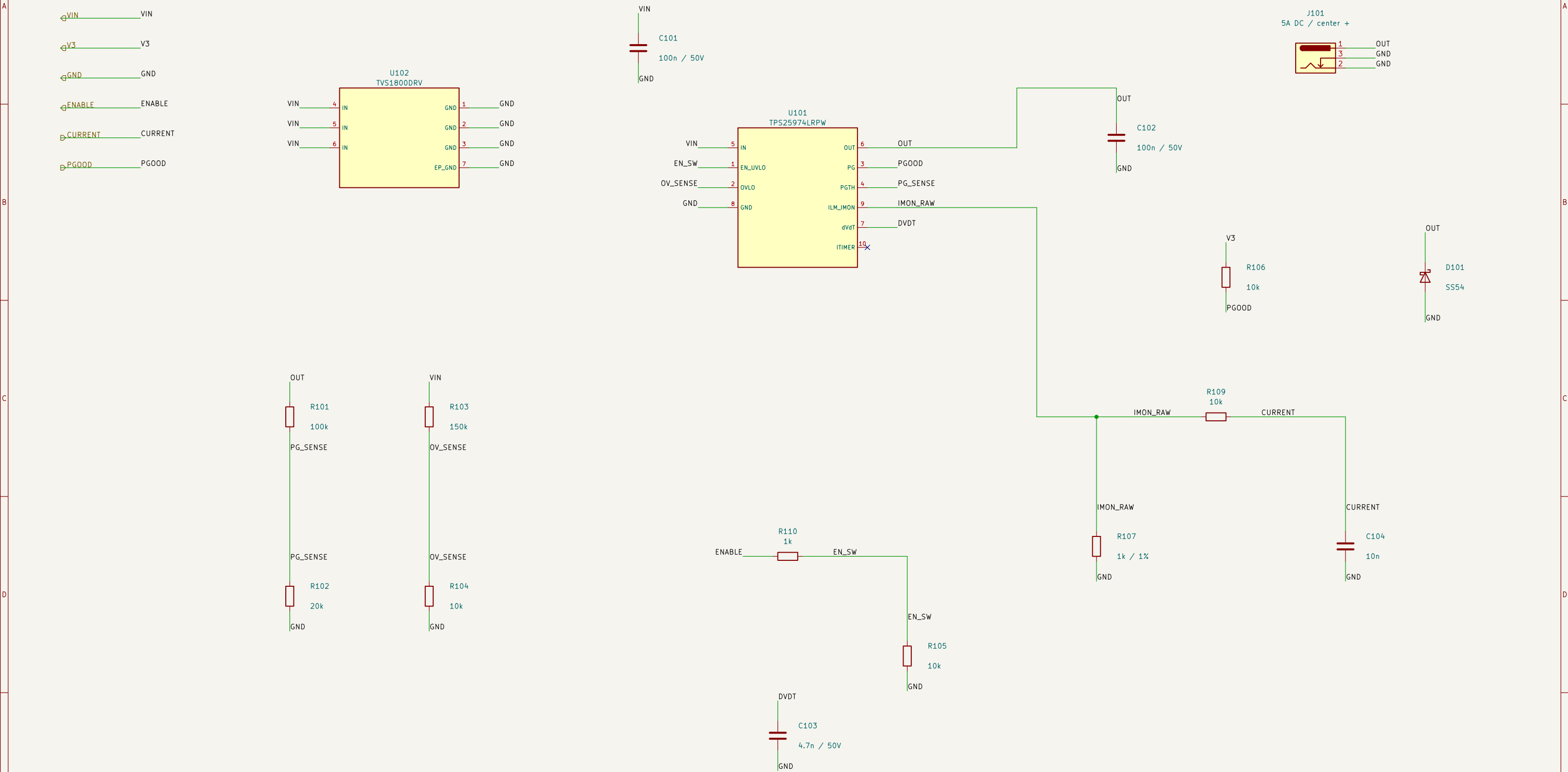


PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44; keep physically away from power components. New firmware pin map is required.

USB HUB | USB2517 | internal port 1 + external ports 2–5; ports 6/7 disabled

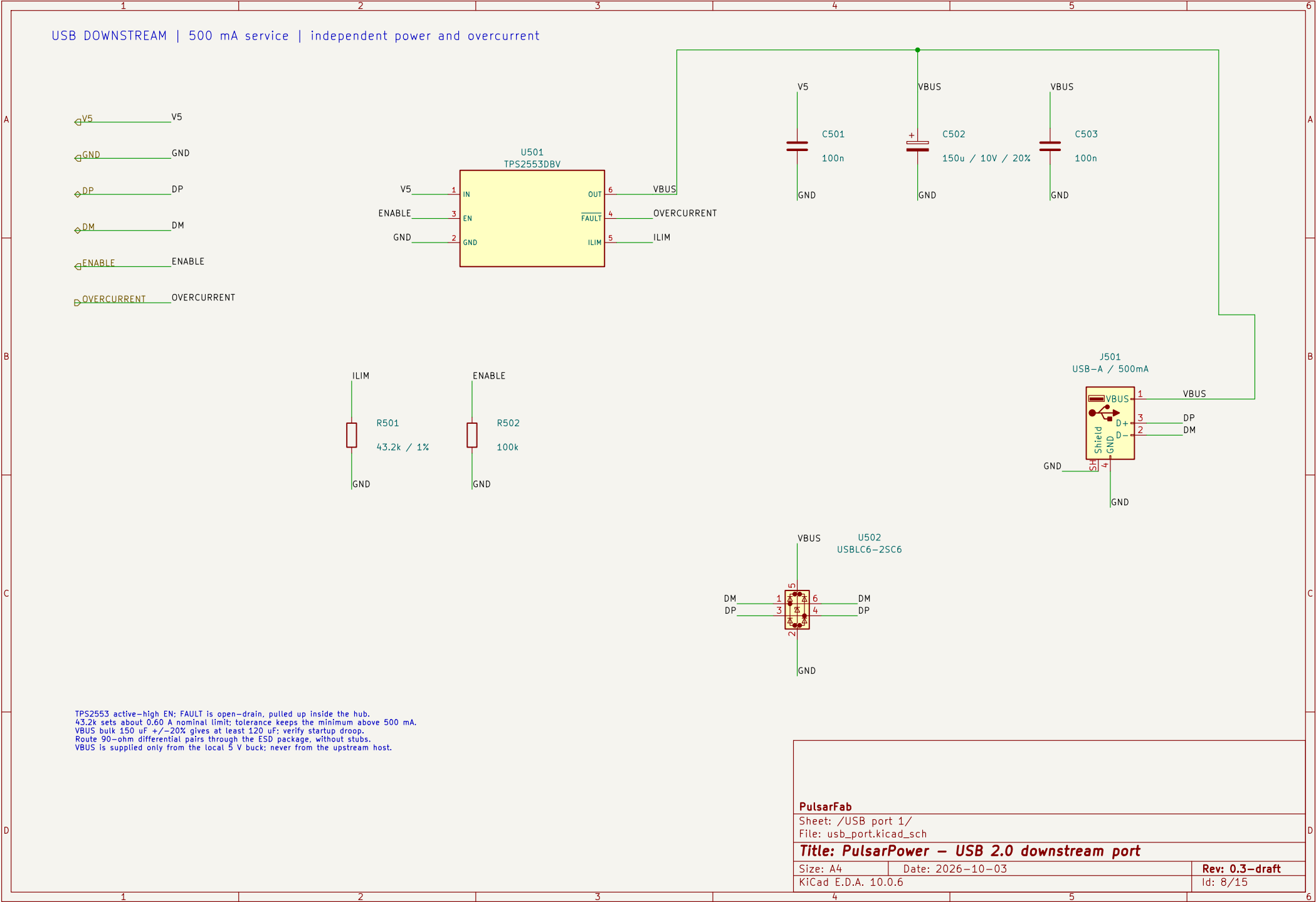


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

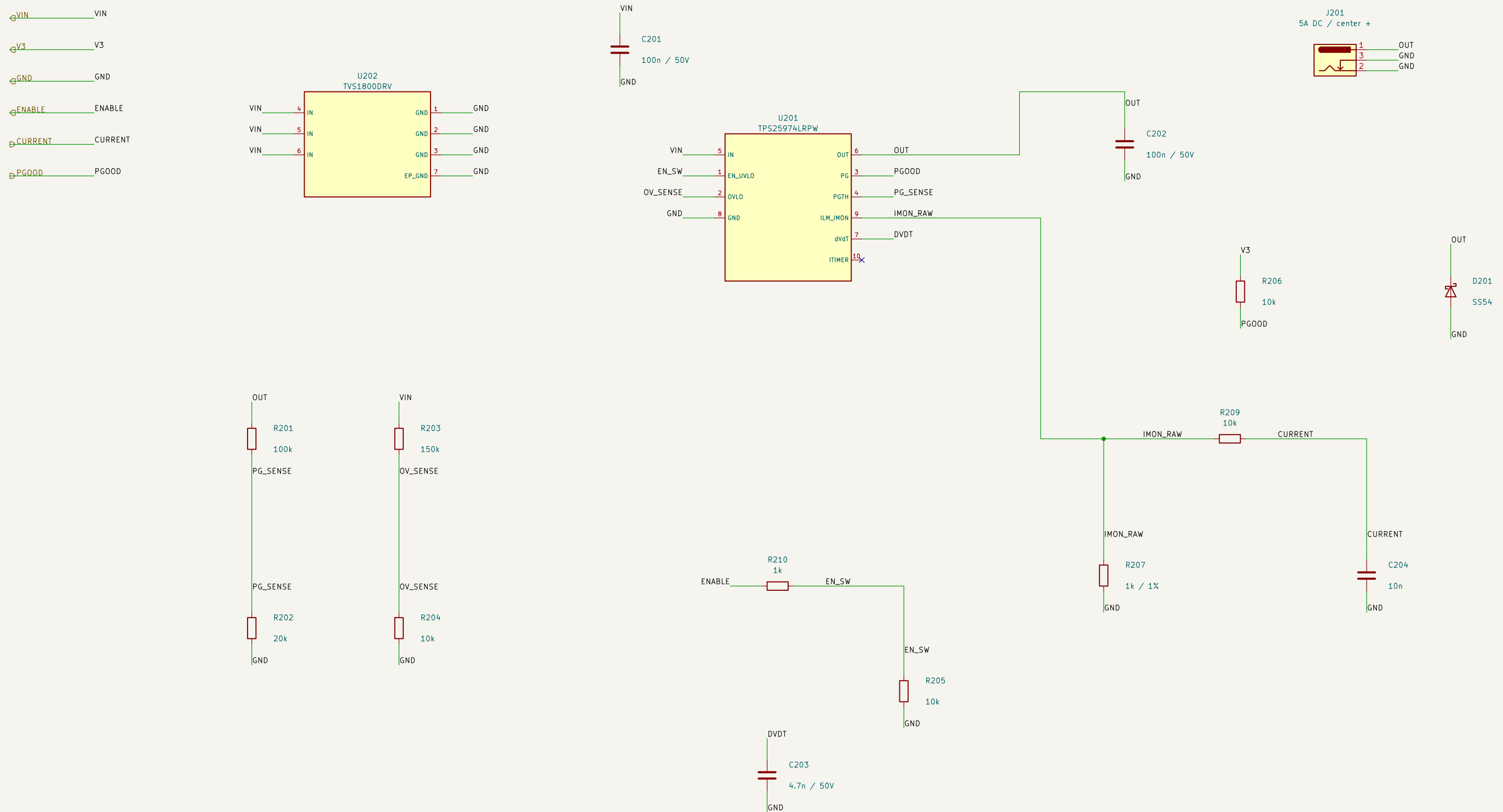


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal.
Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal-via space.
TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

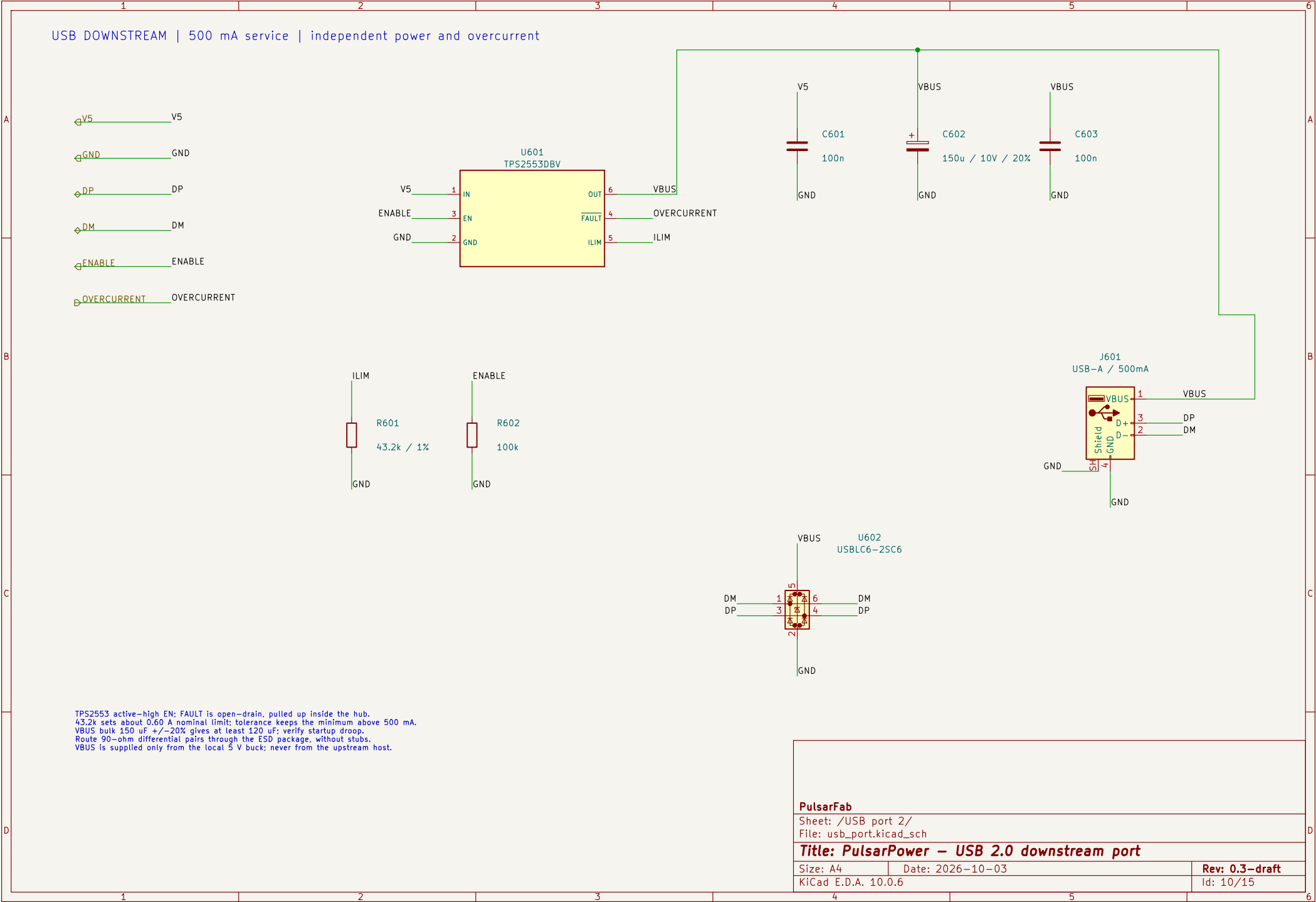


SWITCHED DC OUTLET | 12-18 V / 5 A service | hardware voltage inhibit / latched breaker

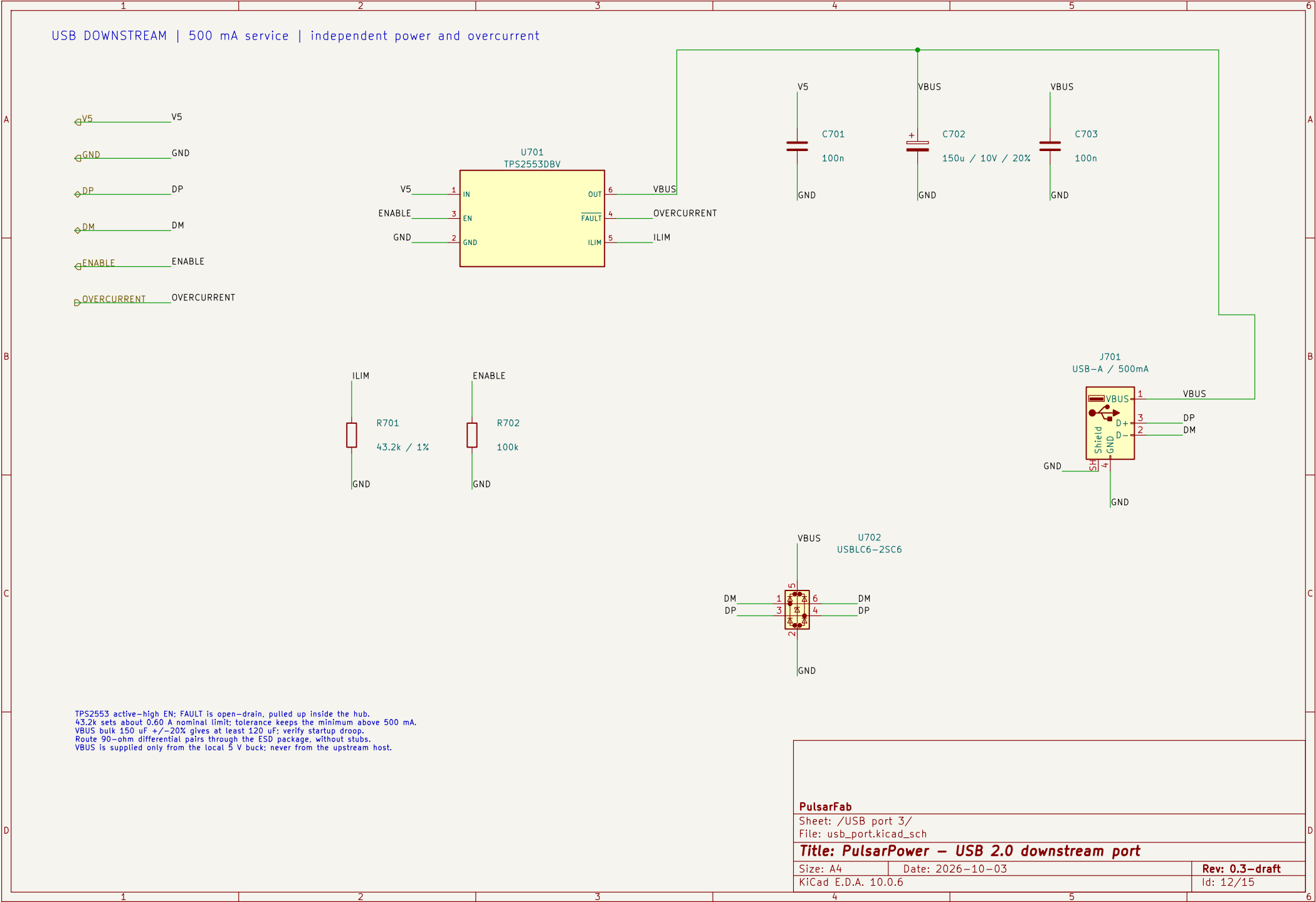


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open; latch-off. ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A. 0.258 V at 5 A. 10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM. Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal. Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs. In pads / OUT pads are power/thermal; NOT GND. Reserve copper and thermal-via space. TVS1800 flat clamp replaces SMBJ18A; 24.7 V max at specified 35 A - 8/20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

Sheet: /USB port 3/
File: usb_port.kicad_sch

Title: PulsarPower – USB 2.0 downstream port

Size: A4

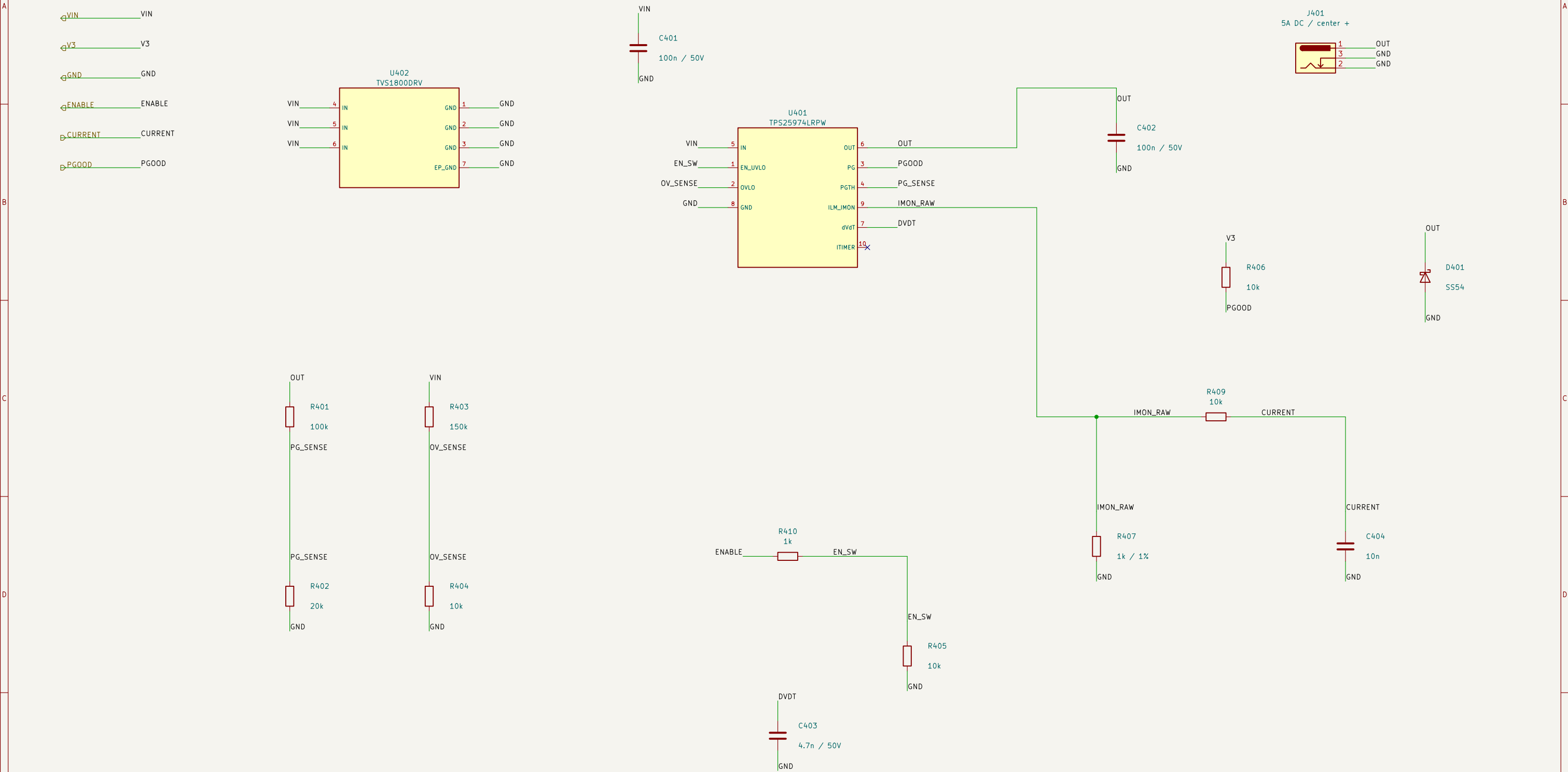
Date: 2026-10-03

Rev: 0.3-draft

KiCad E.D.A. 10.0.6

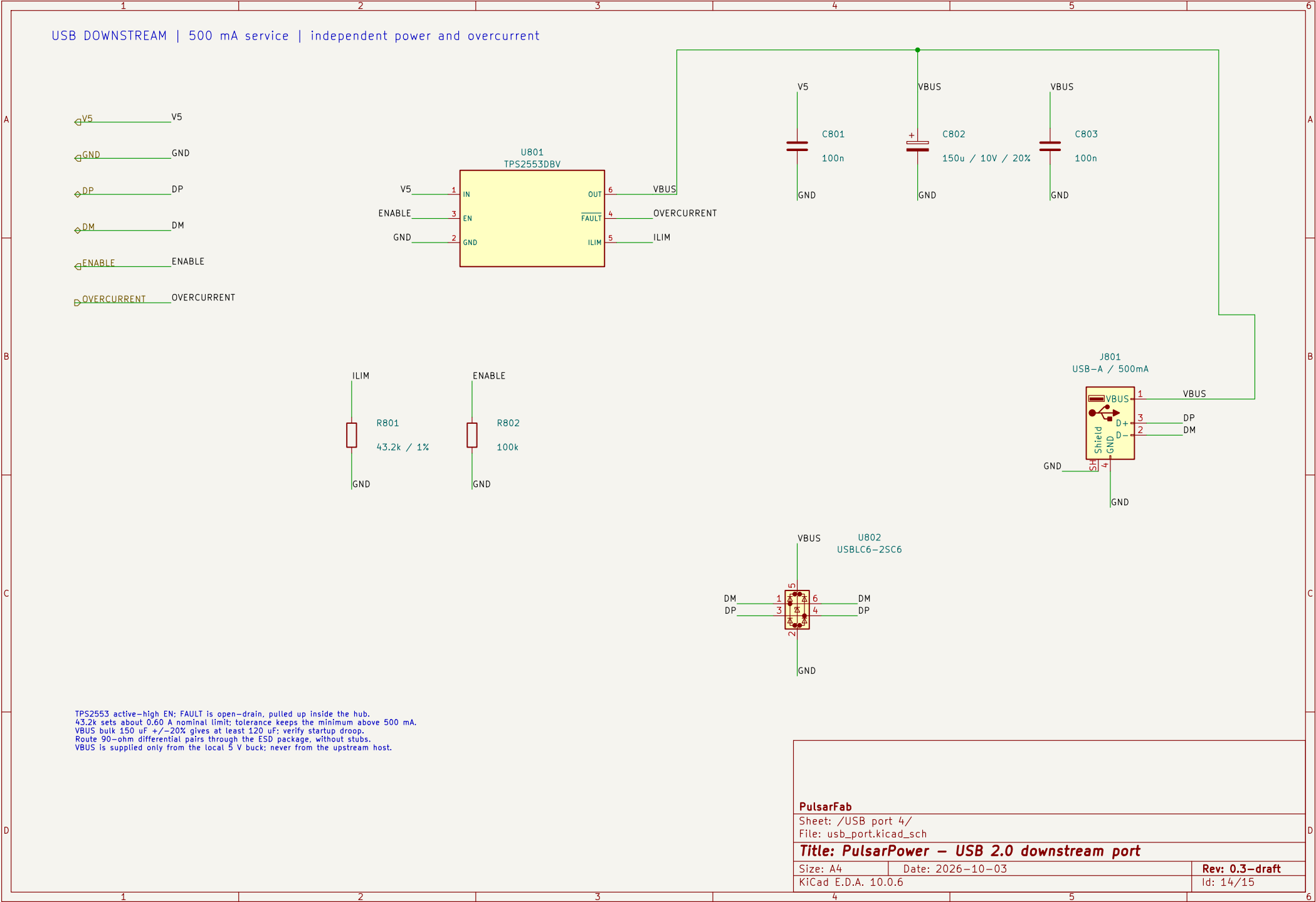
Id: 12/15

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

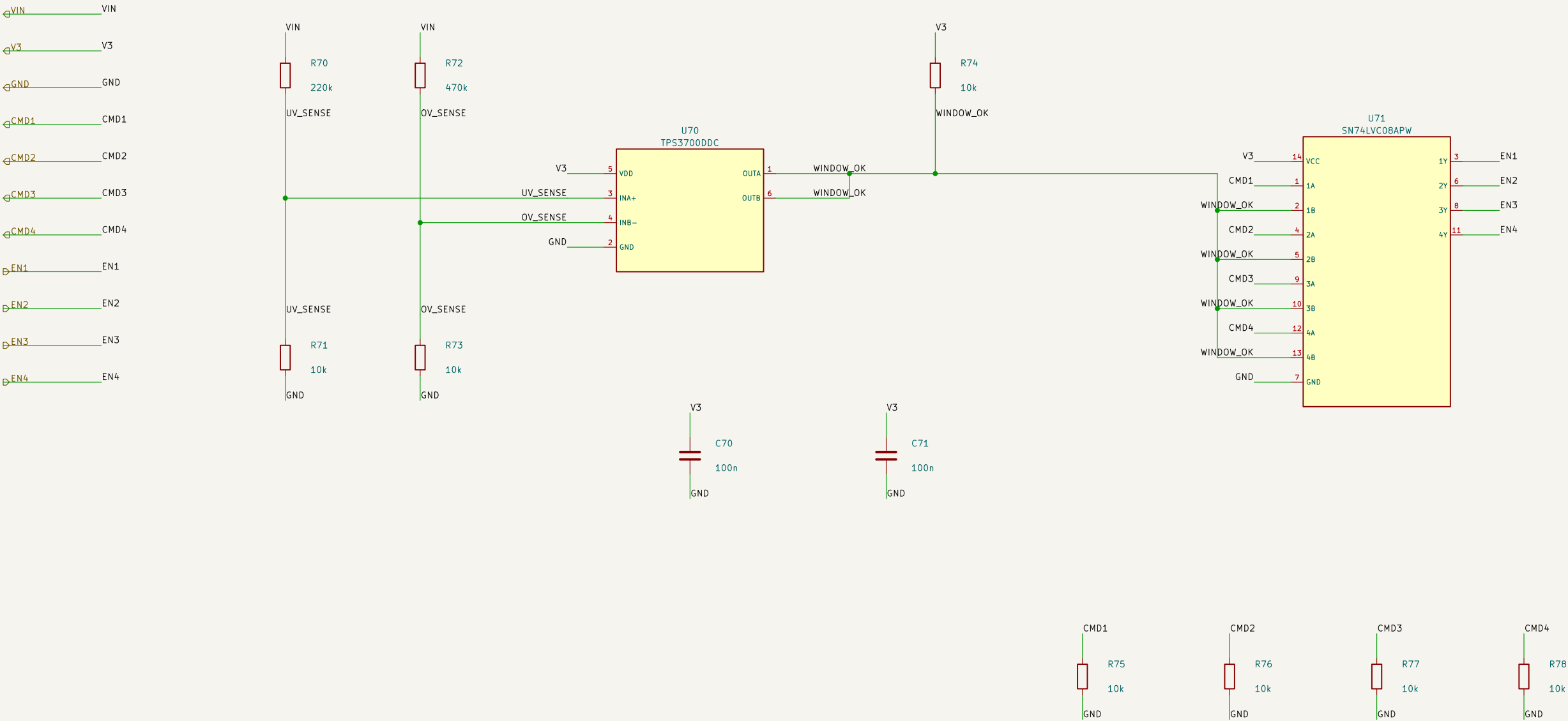


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal.
Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal-via space.
TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



DC PROTECTION | one hardware voltage window | four independent default-off commands



TPS3700: 9.2 V UV rising / 19.2 V OV rising nominal. Open-drain outputs wired together.
74LVC08: ENi = CMDi AND WINDOW_OK; each command has a 10k pulldown.
Each outlet also has an EN pulldown: reset, missing logic supply and disconnected command request OFF.
A voltage recovery can re-enable an asserted command and reset a latched breaker; firmware must deassert on unexpected PG loss.
Shared inhibit is a common failure point, not a redundant safety channel. Local eFuse OVLO remains independent.
This circuit does not disconnect VIN from the regulator, heater stages or eFuse input pins.